

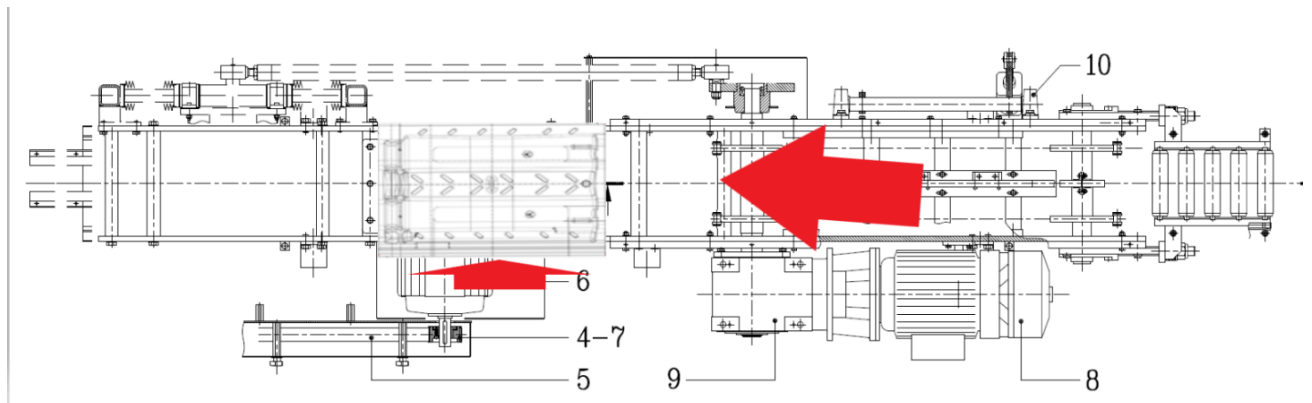
# **ANALYSIS AND SOLUTION FOR MOULD MISALIGNMENT IN THE EXTRUSION BOX 181125**

**Objective:** To identify the root cause and establish the verification and adjustment method to correct the lateral forcing of the moulds against the guides of the extrusion box.

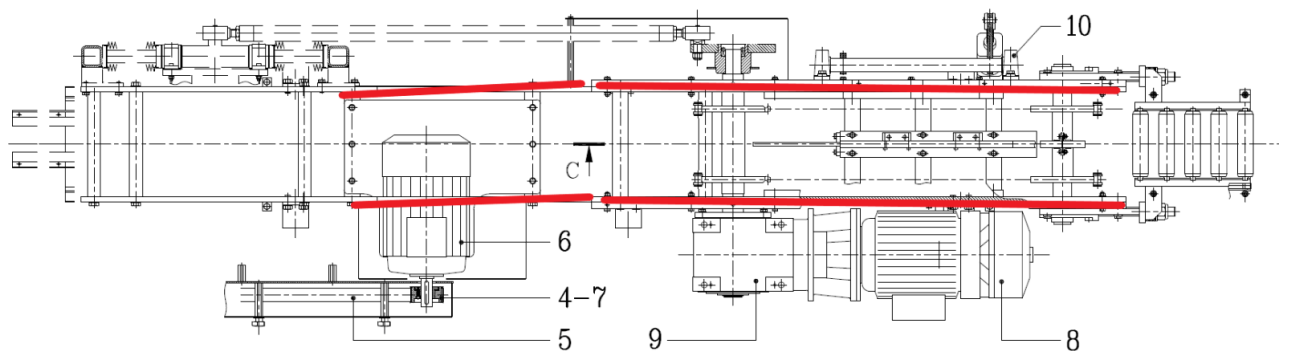
## **1. THEORETICAL ASSESSMENT OF POSSIBLE CAUSES**

Following the initial evaluation, the following hypotheses are proposed as the most probable explanations for the lateral deviation of the moulds:

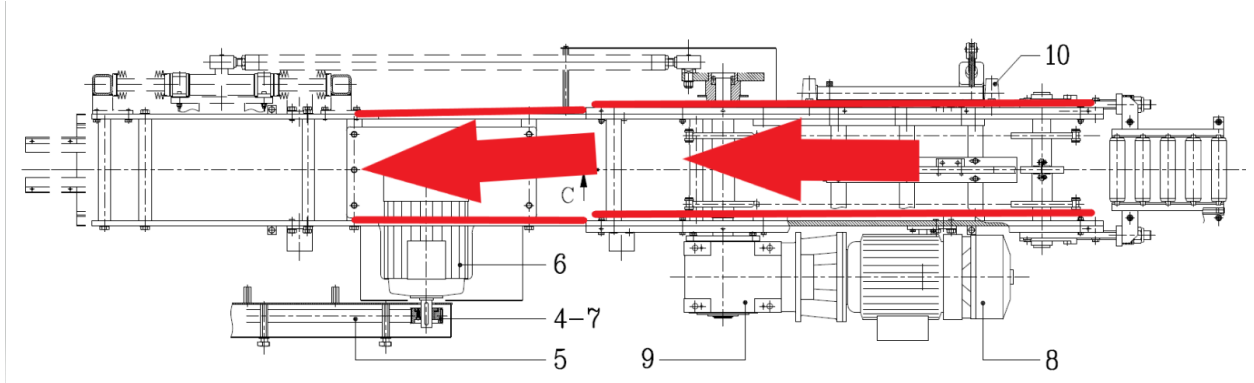
- 1) **Hypothesis A: Misalignment at the Push Point.** The fingers of the push chain trolleys may not be applying force to the center of the mould, but instead pushing it asymmetrically to one side from the start of the cycle.



- 2) **Hypothesis B: General Guide Misalignment.** The guides of the push chain and those of the extrusion box are not on the same plane, creating a progressive deviation at the transition between both sections.



- 3) **Hypothesis C: Displacement of the Push Chain Guides.** The push chain guides themselves may be off-center relative to the theoretical axis of the chain, which would generate an incorrect push before the moulds enters the extruder.



## 2. CHECKS AND VERIFICATION METHODOLOGY

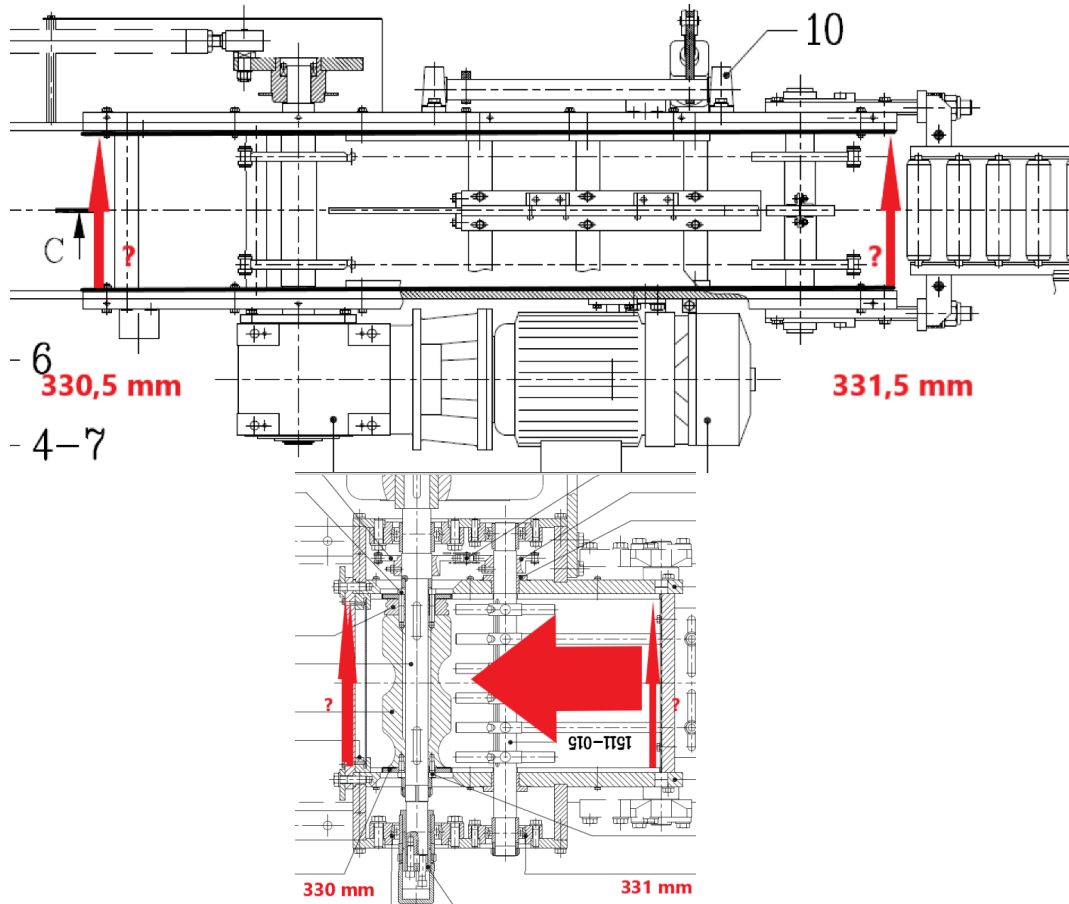
To verify the hypotheses, the following practical checks must be performed:

### a) **Verification of Guide Dimensions and Adjustments**

- **Required Instruments:**
  - **400 mm caliper.**
  - **Inside Caliper** to access and take measurements in spaces where the large caliper cannot be positioned correctly.



- **Procedure:** Measure the width and position of the guides at four critical points: inlet and outlet of the push chain, and inlet and outlet of the extrusion box. This will confirm whether the dimensions are correct and uniform.

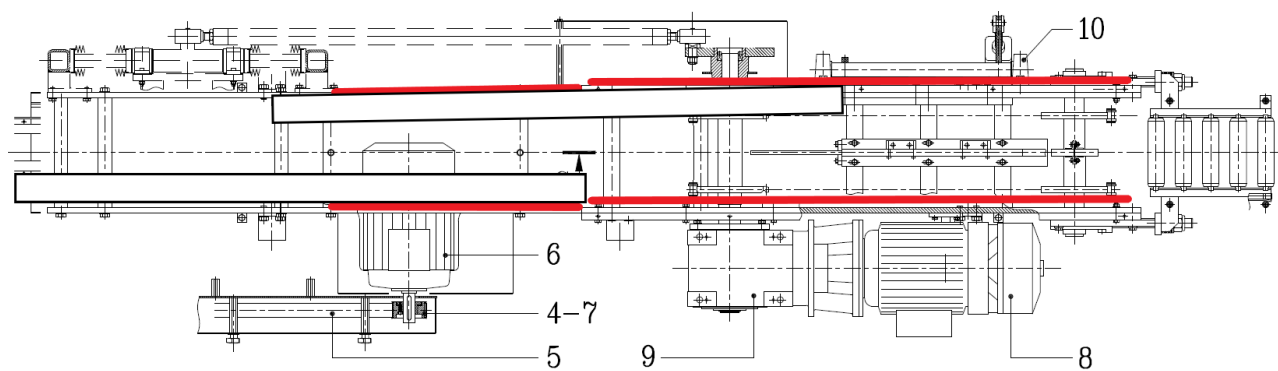
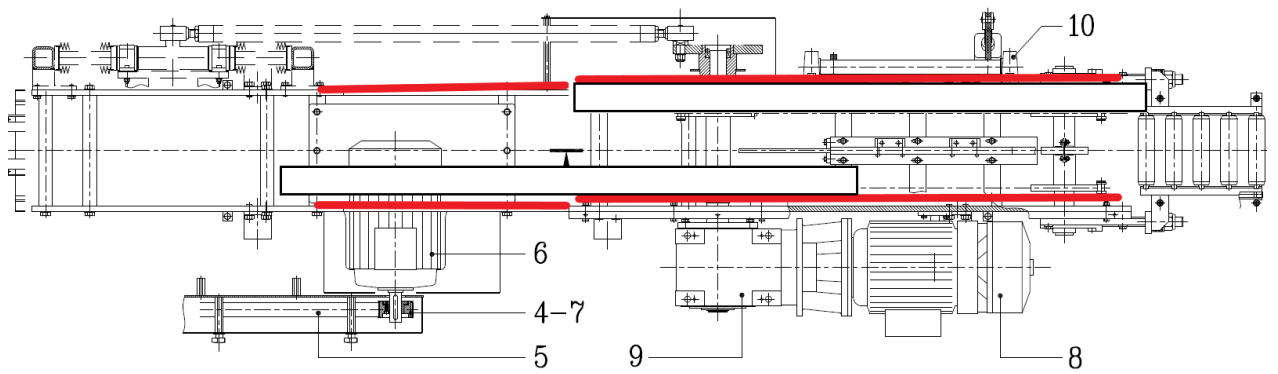


## b) Alignment Check Between Guides

### Required Instrument:

- **2000 mm leveling ruler or straight edge.**
- **Procedure:** Place the ruler longitudinally on the push chain guides and extend it towards the guides of the extrusion box (with the latter open). This will allow for visual and physical verification of the continuity and perfect alignment between both sets of guides, identifying any unevenness or lateral displacement.

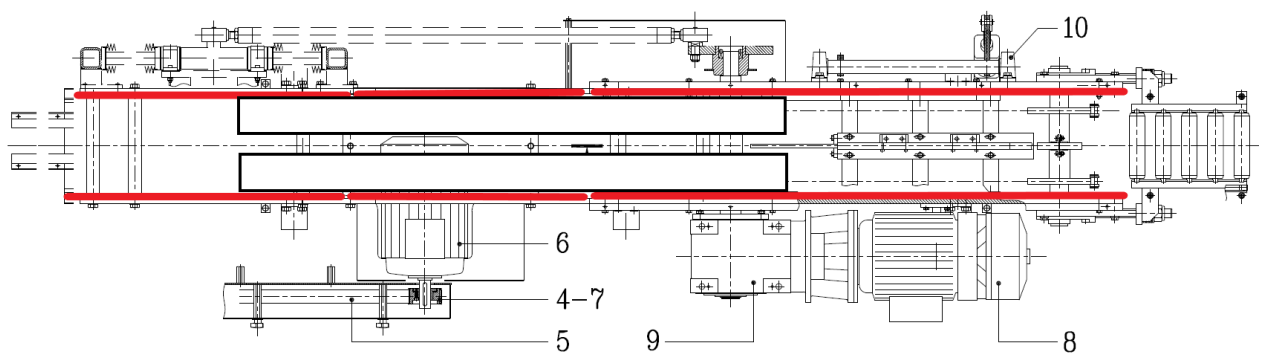




### 3. IDEAL ADJUSTMENT AND SETUP

The goal of the adjustment is to achieve a continuous guide system and a perfectly centered push.

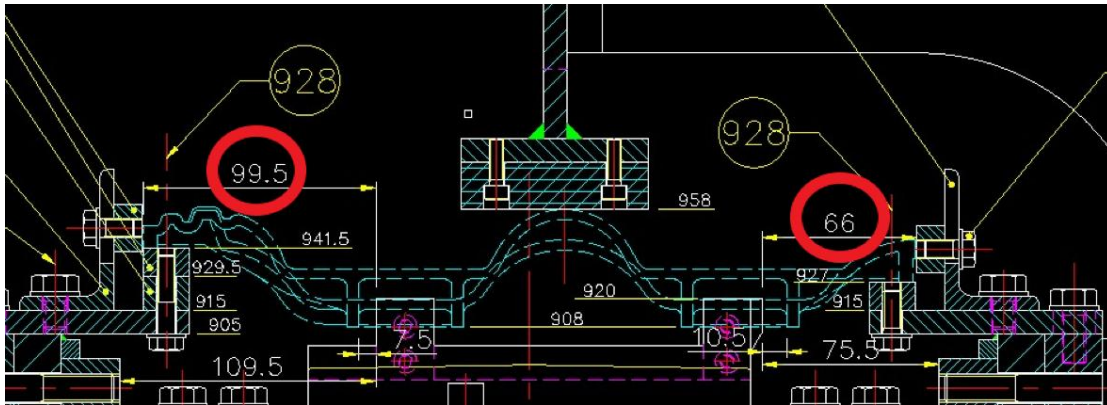
1)**General Alignment:** All guides (push chain, extrusion box, and cutting carriage) must form a perfectly aligned continuous line.



**2)Centering of the Push Fingers:** The fingers of the trolleys must make contact at the theoretical center point of the mould.

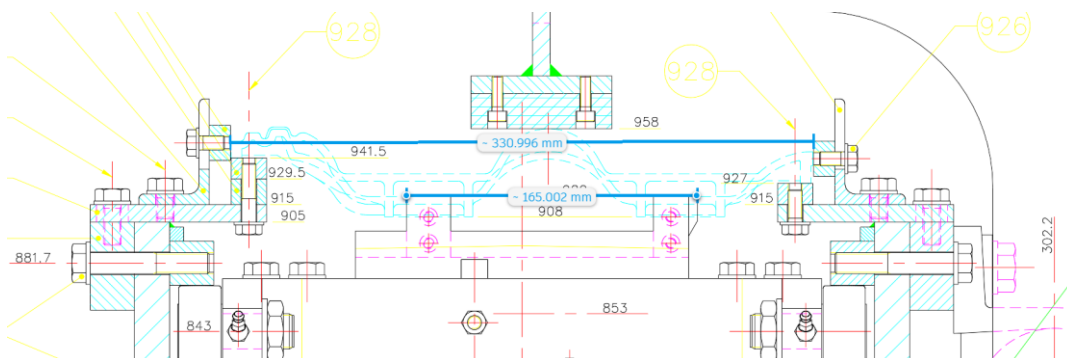
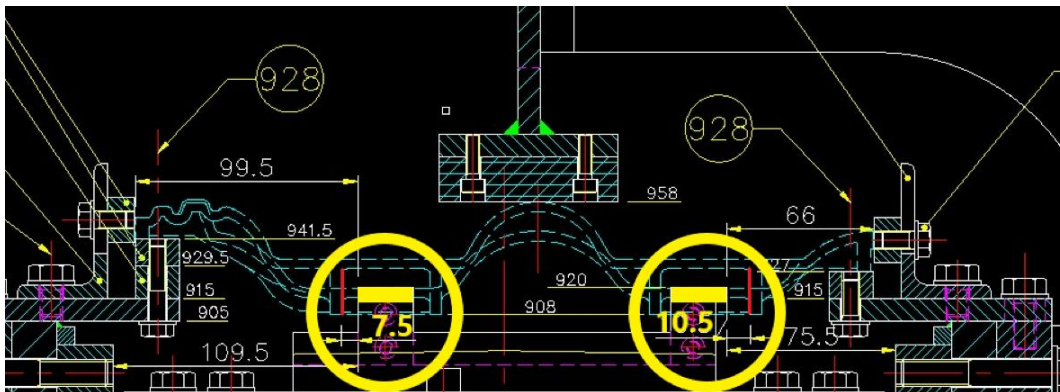
The reference dimensions for this centering are:

- \* **Distance from left finger to left guide: 99.5 mm.**
- \* **Distance from right finger to right guide: 66 mm.**



**3)Lateral Clearance on the Mould:** Once the fingers are centered, the theoretical clearances between the mould and the fingers must be respected:

- \* **Left side clearance: 7.5 mm.**
- \* **Right side clearance: 10.5 mm.**



#### 4. STATIC POSITION AND CLEARANCE VERIFICATION OF PUSH FINGERS

##### **Objective:**

To verify the geometric position and clearances (lateral and upper) of the push fingers in relation to the mould when the system is in its stable working point.

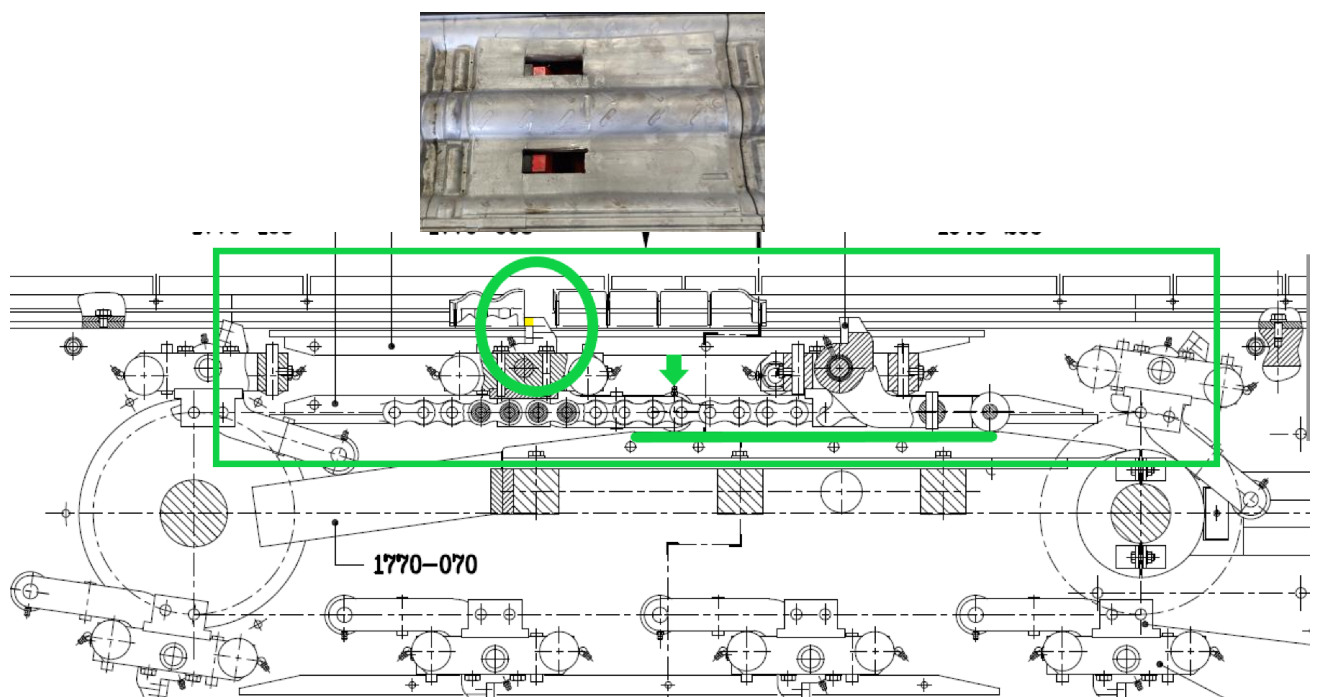
##### **Preparation:**

- 1 test mould, modified with an inspection window on its upper section.



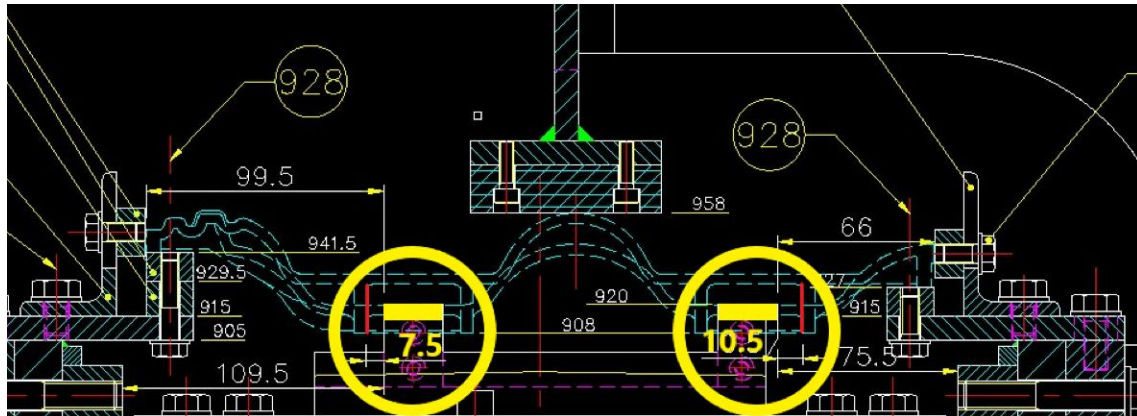
##### **Procedure:**

1. Place the windowed mould on the push chain.
2. Advance the chain until critically positioning the mold: **the roller / bearing of the central trolley must be fully supported on the straight and horizontal section of the central guide.** This position constitutes the reference where pushing is stable and measurements are valid.



### Parameters to Verify:

Once correctly positioned, measure through the inspection window:



- **Lateral clearances:** Space between the flanks of the fingers and the internal walls of the mould.
- **Upper clearance:** Distance between the upper face of the fingers and the mould.
- **Alignment:** Visual verification of the contact point relative to the designated theoretical position.

## 5. DYNAMIC VERIFICATION OF PUSH SEQUENCE

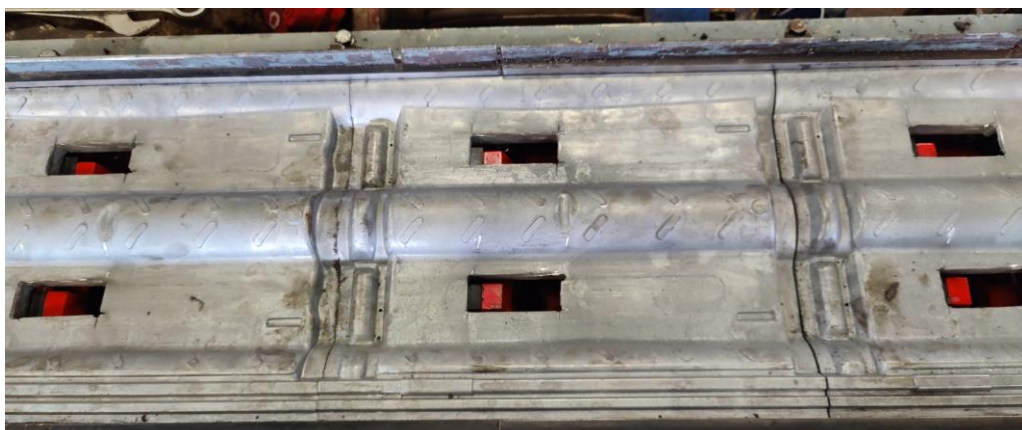
### Objective:

To analyze the sequential behavior and force transition between push fingers of multiple trolleys (3 or 4) during the operational cycle.

### Preparation:

- 3-4 test moulds, unsuitable for production, modified with upper inspection windows.

### Procedure:



1. Arrange the windowed moulds consecutively at the start of the chain.
  2. Pass the set of moulds through simultaneously.
  3. Visually observe, through the windows, the finger interaction in different cycle phases:
    - **Finger under full load:** Exerting the main push on the lower area of the mould.
    - **Finger in disengagement phase:** About to lose contact with the mold it's releasing.
    - **Finger in pre-push phase:** Preparing to initiate contact once the forward finger releases its pressure.
- 

### **GENERAL CONCLUSION:**

The correct system alignment fundamentally depends on the precise positioning of the push chain guides relative to their fingers.

**The static geometric verification (point 4) constitutes the first and most critical step**, establishing the reference working condition.

Once confirmed, this serves as the basis for adjusting the lateral guides of the extrusion box, thereby ensuring integral alignment and operational continuity of the entire line.

The dynamic verification **(point 5)** complements this analysis by validating the correct force transfer during continuous operation.

**Recommendation:** Document findings from both verifications with photographs or sketches of the inspection windows to generate a technical record of the configuration.